

Самки крупных пахучих лягушек (*Odorrana graminea*) предпочитают самцов с более выраженными нелинейными вокальными компонентами.

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Впервые опубликовано: 10 февраля 2022 года

<https://doi.org/10.1002/ece3.8573>

 ПРОСМОТР ПОКАЗАТЕЛЕЙ

Пань Чэнь и Цзиньмэй Ван внесли равный вклад в эту работу и должны считаться соавторами.

Информация о финансировании:

Национальный фонд естественных наук Китая, грант 31872230 (ZF); 31640073 (ZF); ключевая лаборатория провинции Аньхой по сохранению и использованию биологических ресурсов (грант № 591601).



Абстрактный

У бесхвостых земноводных сложность брачных криков может влиять на выбор самки. Согласно результатам текущего исследования, нелинейные явления (НЯВ) могут способствовать усложнению брачных криков и привлечению внимания самок. Результаты недавнего исследования показали, что крики большой вонючей лягушки (*Odorrana graminea*) содержат НЯВ. Однако неизвестно, повышают ли нелинейные компоненты брачных криков *O. graminea* привлекательность самцов. Мы предположили, что самки *O. graminea* будут отдавать предпочтение самцам, издающим звуки с большей долей компонентов, связанных с обработкой естественного языка (P-NLP-C). Чтобы проверить эту гипотезу, мы записали брачные крики 28 самцов и подтвердили, что показатель P-NLP-C положительно коррелирует с размером тела. Мы также измерили размер тела самцов, демонстрирующих амплексус, и самцов, не демонстрирующих амплексус, в естественной среде и обнаружили, что самцы, демонстрирующие амплексус, крупнее самцов, не демонстрирующих амплексус. Кроме того, результаты экспериментов с выбором между самцами, демонстрирующими и не демонстрирующими амплексус, показали, что самки предпочитают самцов с более крупным телом. Кроме того, эксперименты по изучению фонотаксиса показали, что самки предпочитают самцов, издающих звуки с высоким показателем P-NLP-C. Результаты показывают, что высокий

показатель P-NLP-C в звуках может повышать привлекательность самцов, а показатель P-NLP-C может служить для самок *O. graminea* ключевым источником информации о состоянии тела самца. Наше исследование позволяет по-новому взглянуть на роль NLP в выборе партнера среди бесхвостых земноводных.

1 ВВЕДЕНИЕ

Звуковая коммуникация играет важную роль в выборе самки, особенно у бесхвостых земноводных (Наринс и др., 2006; Чжан и др., 2015). Характеристики брачных криков самцов, особенно их сложность, являются важными факторами, которые учитывают беременные самки при выборе партнера (Куи и др., 2016; Райан и др., 2019). Самки обращают внимание на крики самцов и отдают предпочтение сложным крикам перед простыми (Гриди-Папп и др., 2006). Акустическая сложность часто связана с вариациями спектрально-временных характеристик (например, частоты, продолжительности и количества слогов) вокальных сигналов (Фи и др., 1998; Маркес, 1995; Райан, 1983). У некоторых видов, издающих нелинейные звуки, улучшение содержания нелинейных феноменов (НФ) является важным способом повышения сложности издаваемых звуков (Fitch et al., 2002; Rice et al., 2011; Wu et al., 2021).

Нелинейные акустические явления, как правило, имеют сложную структуру (Rebout et al., 2020). Две асимметричные голосовые связки с разной массой или длиной вызывают нелинейные явления (NLP), которые включают в себя четыре акустические характеристики: скачки частоты, субгармоники, детерминированный хаос и бифоникацию (Fitch et al., 2002). НЛП, как отличительная структурная особенность акустических сигналов, широко распространено среди позвоночных (Дигби и др., 2014; Фенг, Риде и др., 2009; Фитч и др., 2002; Райс и др., 2011; Чжан и др., 2017). Частота и интенсивность нейронно-сетевых паттернов в вокализации не только повышают индивидуальность звуков, но и привлекают больше внимания со стороны реципиентов при индивидуальном взаимодействии, что было неоднократно подтверждено на примере рыб и млекопитающих (Fitch et al., 2002; Володина и др., 2006; Уайлден и др., 1998). Ряд исследований был посвящен общим функциям нейролингвистического программирования, например в сигналах тревоги (Blumstein & Recapet, 2009; Townsend & Manser, 2011) и передаче информации о направлении движения и личности (Digby et al., 2014; Kaplan et al., 2018), но лишь немногие из них были посвящены выбору партнера для спаривания, особенно у лягушек (Wu et al., 2021).

The large odorous frog (*Odorrana graminea*) can produce calls that contain four NLP components (subharmonics, deterministic chaos, frequency jumps, or biphonations), and 84.6% of the vocalizations contain one or more nonlinear components (Zhang et al., 2021). *O. graminea* is an arboreal species inhabiting areas around loud streams and waterfalls in select regions of China (Chen, 1991). Similar to that of a sympatric species, *Odorrana tormota*, the calls of *O. graminea* consist of ultrasonic and NLP components (Shen et al.,

2011; Zhang et al., 2021). In *O. tormota*, smaller males with higher call frequencies had greater mating success than larger rivals, as their calls were more conspicuous in the species' habitat with intense but predominantly low-frequency stream noise (Zhang et al., 2020). Whether the mode of mate choice in *O. graminea* is similar to that in *O. tormota* remains unknown. Furthermore, since NLP components can improve the complexity of courtship calls, knowledge about the roles of NLP levels in attracting mates is essential for understanding mate choice as a social function, but studies on the roles of NLP levels in mate attraction have been conducted in only one anuran species (*O. tormota*) that inhabits subtropical stream areas (Wu et al., 2021). Therefore, it is unclear whether or how the nonlinear components of courtship calls improve male *O. graminea* attractiveness and whether females of *O. graminea* prefer males with a higher proportion of NLP components (P-NLP-C). To address these issues, we investigated (1) the mode of mate choice in *O. graminea* and (2) the effects of P-NLP-C on mate choice.

2 METHODS

2.1 Study site and subjects

Field observations and laboratory experiments involving *O. graminea* were conducted at Fu Creek in Huangshan, Anhui Province, China (118°08'44.89"E, 30°05'01.61"N, elevation: 600 m asl). Fieldwork at the location over the past seven years has revealed that the peak month for breeding activity in *O. graminea* is May. As mate choice may vary with different periods in the breeding season, field and laboratory studies were carried out during peak breeding season, from May 2 to May 28 in 2019 and 2020, to minimize sampling bias. Each night, we captured as many amplexed pairs as possible during the peak reproductive hours (between 19:30 and 22:30; Figure 1). All captured individuals were placed in containers with water and stones in time and then brought back to the test chamber. The local nightly temperature and humidity during the study period ranged from 19 to 24°C and 90% to 96%, respectively. The test chamber was located near the sampling site and maintained a similar temperature, humidity, and dark light environment. Each individual was in the test chamber for no more than three days and released back to the sampling site immediately after testing. To avoid repeated capture, each frog was given a finger cutting mark before returning to its habitat. All the behavioral observations conducted in this study were performed in accordance with the current laws of the China and the Animal Care and Use Committee at the Anhui Normal University (Permit # 00111).



FIGURE 1

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An amplexed pair of *Odorrana graminea* in the field. Photo by Jinmei Wang

2.2 Field sampling

We captured 30 amplexed pairs that were found on top of boulders in the middle of the mountain stream and in bamboo groves next to small waterfalls in the stream. Three to six solo males (i.e., non-amplexant) in the vicinity (< 0.5 m) of an amplexed pair were randomly captured and housed in separate terrariums ($22 \times 17 \times 15$ cm). Using a caliper (Spi2000 Wiha, Germany) with an accuracy of ± 0.1 mm, we measured the snout-vent lengths (SVLs) of the frogs. The SVL data of the amplexant males were later compared to those of solo males. The number of amplexed pairs we spotted and captured each night varied from 0 to 5. No amplexed pair was found in the field when the ambient temperature was below ~ 15 °C. A total of 123 solo males were captured, and their characteristics were compared with those of amplexed males in the field (Table 1).

TABLE 1. The SVLs of amplexant and non-amplexant males from nightly field observation. Shown are the mean $\pm$ SD

Night #	Mean SVL of amplexant males (mm)	Mean SVL of non-amplexant males (mm)
1#	50.45 $\pm$ 2.77 (N = 4)	48.52 $\pm$ 2.48 (N = 16)
2#	50.52 $\pm$ 2.52 (N = 5)	49.85 $\pm$ 2.19 (N = 21)
3#	49.90 $\pm$ 3.10 (N = 5)	48.82 $\pm$ 2.18 (N = 19)
4#	50.75 $\pm$ 2.10 (N = 4)	49.54 $\pm$ 2.20 (N = 16)
5#	50.15 $\pm$ 1.20 (N = 2)	49.07 $\pm$ 2.23 (N = 9)
6#	51.30 $\pm$ 0.57 (N = 2)	48.79 $\pm$ 3.20 (N = 11)
7#	51.15 $\pm$ 1.20 (N = 2)	49.93 $\pm$ 2.41 (N = 9)
8#	51.70 $\pm$ 0.57 (N = 2)	48.30 $\pm$ 3.65 (N = 8)
9#	51.25 $\pm$ 1.48 (N = 2)	47.36 $\pm$ 2.37 (N = 8)
10#	51.20 $\pm$ 0.42 (N = 2)	49.83 $\pm$ 2.12 (N = 6)
Total	50.74 $\pm$ 2.01 (N = 30)	49.07 $\pm$ 2.48 (N = 123)

2.3 Audio recording and call analysis

To quantitatively analyze the relationship between male body size and the P-NLP-Cs in vocalizations, we recorded the vocalizations of 28 actively calling males (not in amplexus) for three evenings under similar ambient conditions (temperature: 20 ~ 22°C, humidity: 92% ~ 96%, ambient noise: 70 ~ 78 dB SPL peak) in the field and measured their SVLs. Male calls were recorded by using a digital audio recorder (Sound Devices 702, Sound Devices, WI, USA, frequency range: 10 Hz ~ 96 kHz) with a sampling rate of 96 kHz and 16-bit accuracy and a miniature omnidirectional condenser microphone with a flat frequency response over 20 ~ 20,000 Hz (AKG model C417, AKG Acoustics, Vienna, Austria) (Zhang et al., 2015). For each male, we recorded at least 8 calls.

Calls were initially analyzed with SELENA, a custom-designed program (Feng, Riede, et al., 2009), to produce narrow-band spectrograms and determine the call durations. Next, calls comprising multiple notes, the duration of individual notes, the duration of signal breaks, and the overall call duration were measured. Then, different temporal segments of each call were identified with PRAAT based on visual inspection of the narrow-band spectrogram (Figure 2; Boersma & Weenink, 2007). After the call segments were finished, the time of occurrence of each segment was noted. In addition, the durations of the various NLP segments (i.e., subharmonics, deterministic chaos, frequency jumps, biphonations) were

measured. Finally, these durations were calculated as percentages of the total call duration, and the average NLP percentage for each male frog was calculated. The fundamental frequency (f_0) was tracked for each harmonic segment using the “pitch tracking” mode in PRAAT, with a 1-ms interval.

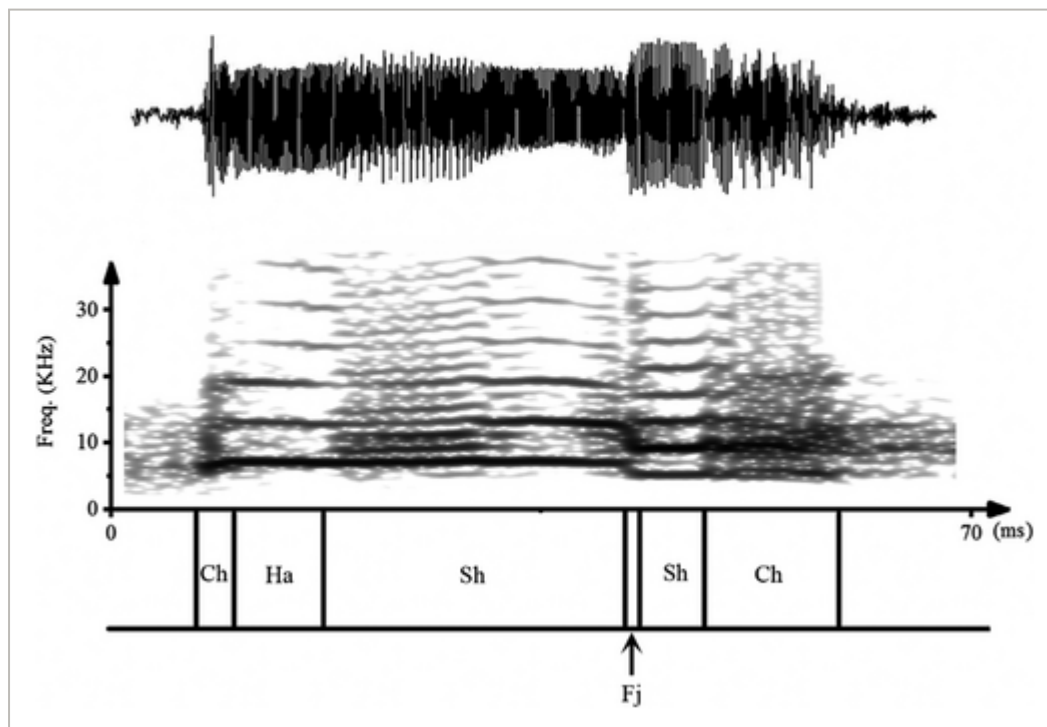


FIGURE 2

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Temporal segmentation of calls. Shown here is the waveform (top trace) and spectrogram (bottom trace) of a call of male *Odorrana graminea*. The call was segmented into segments containing harmonic (Ha), subharmonics (Sh), deterministic chaos (Ch), and frequency jump (Fj). This particular call does not show biphonation segments

2.4 Amplexus experiments

To determine the male body size preferred by females, we performed two different two-choice amplexus experiments. The behaviors of the males and females were recorded by a camcorder (Sony model HF M40). In the first two-choice experiment (Experiment A), we paired one gravid female with two randomly captured males with varying body sizes in a test terrarium (38 × 38 × 28 cm) for up to 15 min. We placed the female in the center of the test terrarium and the two males equidistant from the female. If the pairing did not result in amplexus within 15 min, namely, the female rejected both males, we substituted two new males and repeated the experiment (at most one repeat). Upon amplexus, we separated the pair and measured the SVLs of the amplexant male and non-amplexant male. In total, 27 out of 30 experiments resulted in amplexus.

In the second two-choice experiment (Experiment B), a female was paired with two males that had been previously paired with a different female. The goal was to determine whether a particular male phenotype was preferred by different females. To execute the second two-choice experiment, we removed the female previously paired (in Experiment A) from the test terrarium and allowed the males to remain and rest in the terrarium for at least 30 minutes. Then, we placed another female into the terrarium and began the second two-choice experiment (Experiment B). All individuals in each experiment were randomly assigned and never tested more than once in the same experiment.

2.5 Phonotaxis experiments

Three acoustic-stimulus pairs were synthesized using Cool Edit Pro 2.1 software based on the advertisement calls of different males that have been shown to attract females. We first chose some typical calls that fits the criteria (specific frequency or P-NLP-C), then synthesized stimulus from different call components in a certain ratio, and finally standardized some parameters including the call duration and amplitude. The characteristics of stimulus were based on the f_0 and P-NLP-C of the calls observed in this and previous studies (Zhang et al., 2021). In phonotaxis test 1, the goal was to test the effect of frequency variation on female choice while the P-NLP-C at a constant low level. The f_0 of the two stimulus were 2600 Hz, representing low-frequency calls (indicated by L f_0), and 7500 Hz, representing high-frequency calls (indicated by H f_0), while the differences in the temporal and P-NLP-C characteristics were no more than 20 ms and 3%, respectively (Figure 3a,b). In phonotaxis test 2, the goal was to test the effect of P-NLP-C variation on female choice while the f_0 at a constant normal level. The P-NLP-C in the two stimulus were 10% for calls with a low P-NLP-C (indicated by L1-NLP) and 50% for calls with a high P-NLP-C (indicated by H1-NLP), while the differences in the temporal and f_0 characteristics were no more than 20 ms and 100 Hz, respectively (Figure 3c,d). In phonotaxis test 3, the goal was to test the consistent of female choice for the P-NLP-C level of calls compared to test 2 while adjusting the P-NLP-C and f_0 in stimulus. The P-NLP-C in the two stimulus were 20%, representing calls with a low P-NLP-C (indicated by L2-NLP), and 75%, representing calls with a high P-NLP-C (indicated by H2-NLP), while the differences in the temporal and f_0 (high level) characteristics were no more than 20 ms and 100 Hz, respectively (Figure 3e,f). Stimulus pairs were constructed as follows: (1) L f_0 versus H f_0 , (2) L1-NLP versus H1-NLP, and (3) L2-NLP versus H2-NLP.

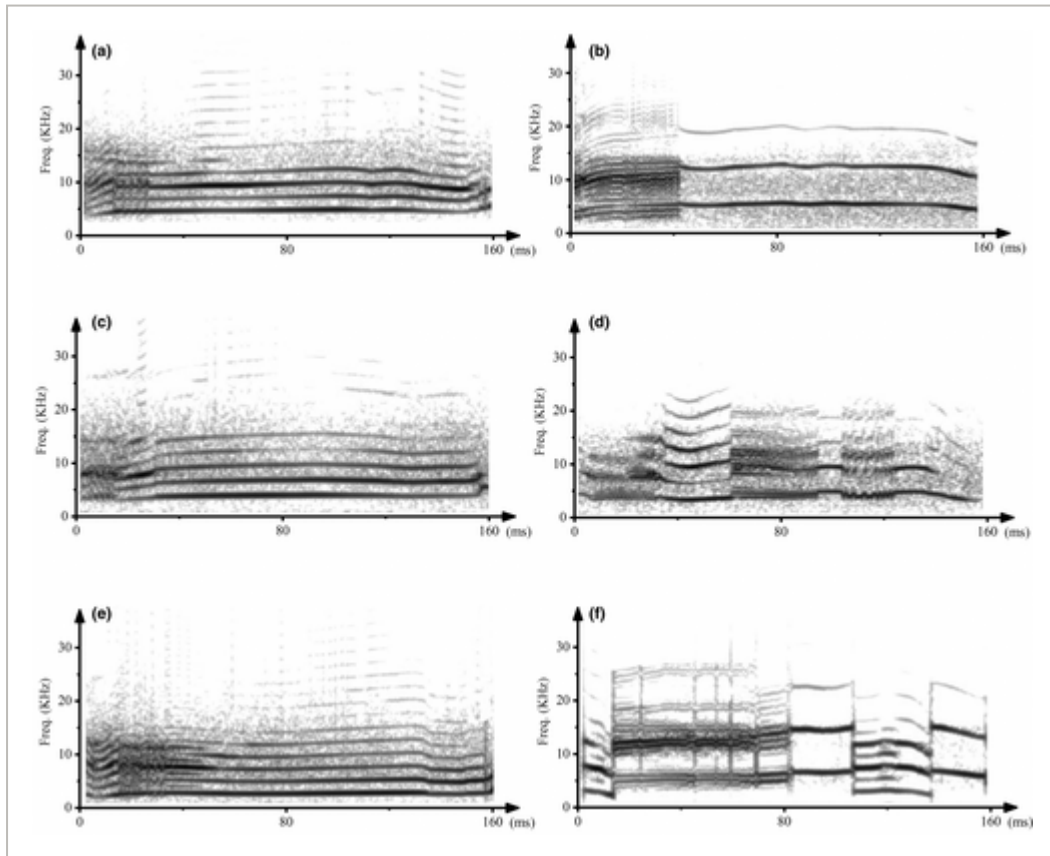


FIGURE 3

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Spectrogram of stimuli pairs. (a) (NLP: 10%; f_0 : 2600 Hz) and (b) (NLP: 13%; f_0 : 7500 Hz) are a stimuli pair for phonotaxis test 1. (c) (NLP: 10%; f_0 : 3800 Hz) and (d) (NLP: 50%; f_0 : 3900 Hz) are a stimuli pair for phonotaxis test 2. (e) (NLP: 20%; f_0 : 4900 Hz) and (f) (NLP: 75%; f_0 : 5000 Hz) are a stimuli pair for phonotaxis test 3

We performed the phonotaxis tests in a hemi-anechoic chamber (L × W × H: 200 × 120 × 150 cm) in which the walls and ceiling were covered by sound-absorbing cotton (Figure 7a). Moreover, we covered the floor and loudspeakers (Altec Lansing Orbit iM227) with branches and grass and misted water into the air to simulate humidity. For each test, we placed a female under an inverted, acoustically transparent experimental box (27 × 17 × 15 cm) on the floor of the arena at a centrally located release location. We moved this box with a string from outside the chamber to grant the female access to the entire arena, which was illuminated by infrared light. During the tests, the behaviors of females were recorded with a camcorder (Sony model HF40) located outside the chamber.

We broadcasted acoustic stimuli every 15 s (2 stimulus calls per min per loudspeaker, simulating the natural call rate) using loudspeakers on the floor of the chamber. To avoid the potential side biases, we randomized the loudspeaker assignments for each stimulus pair (Zhao et al., 2017). The distance between each loudspeaker and the release location was 100 cm, and both loudspeakers were balanced to provide 80 ± 2 dB SPL at the release location. At the start of the experiment, females were stimulated with a 1-min natural

chorus recording using two loudspeakers. Then, we broadcasted the stimuli for approximately 1 min before lifting the experimental box. A choice was confirmed when a female frog approached within 20 cm of a loudspeaker without simply following the wall. To determine whether a particular call characteristic was consistently chosen by the same female, we performed the second phonotaxis experiments using the same females with different positions of the designated target loudspeakers (females were subjected to a rest period of at least 2 h prior to the next test). Next, the time to leave (the time from removing the box until the female exited the release location) and the time to choice (the time from exiting the release location to reaching the choice boundary) were recorded in all experiments, and the averaged phonotaxis time from the two trials was considered the final result. If a female did not approach any loudspeaker within 10 min, we reran the test after a 30-minute rest period. If the female still failed to approach a speaker, no choice was recorded (Zhu et al., 2017). Six females were found to be about to oviposit before the experiment and released back to their natural habitat in time. In total, 24 gravid females were used in the phonotaxis tests. The order of test 1 and test 2 was random, but test 3 was performed after test 2. The order of all individual in each experiment was random.

2.6 Statistics

All statistical analyses were conducted using SPSS 24.0 (IBM SPSS Inc). For all pooled data, we performed the Shapiro–Wilk test and Levene test to determine whether the data were normally distributed and exhibited homogeneous variance, respectively. For data that met these criteria, we performed a parametric test; otherwise, a nonparametric test was used. The Mann–Whitney U test was used to analyze differences in body length parameters between amplexant males and solo males in the field, while the paired t test was performed to analyze differences in body lengths between amplexant males and non-amplexant males in the amplexus experiment. Linear-regression analysis was used to analyze body length correlations between male and female frogs in the amplexus test. In addition, this analysis was also performed to analyze correlations of male body length with the proportions of NLP and the fundamental frequency in male vocalizations. The binomial test was used to evaluate the phonotaxis data for females responding to male calls with different f_0 and P-NLP-C. The Mann–Whitney U test was used to compare female phonotaxis time between different stimulus pairs in three phonotaxis experiments.

3 RESULTS

3.1 Female mate choice in the field

The field observations revealed that on average, amplexant males had a larger SVL (mean SVL $\pm$ SD = 50.74 $\pm$ 2.01 mm, N = 30) than non-amplexant males (mean SVL $\pm$ SD = 49.07 $\pm$ 2.48 mm, N = 123), as shown in Figure 4a. The difference was statistically significant (Mann–Whitney U test, p < .05). The pooled data were consistent with the data

collected nightly. Table 1 shows that the average SVL of amplexant males was larger than that of solo non-amplexant males for all 10 nights for which we located amplexed pairs. The SVL of males (mean SVL $\pm$ SD = 50.74 $\pm$ 2.01 mm, N = 30) and females (mean SVL $\pm$ SD = 97.72 $\pm$ 3.52 mm, N = 30) in amplexed pairs was not significantly correlated (linear-regression analysis, R^2 = 0.000, p = 1.000, Figure 5). Thus, *O. graminea* showed no size-assortative mating.

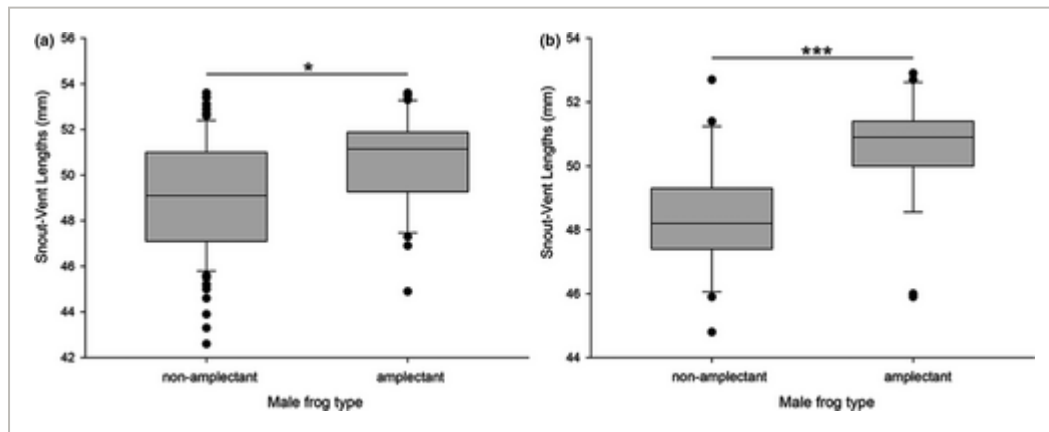


FIGURE 4

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Differences of snout-vent lengths (SVLs) of amplexant males and non-amplexant males. (a) Measurements from males derived from field observations (N = 30 for amplexant males, N = 123 for solo males, Mann-Whitney U test, $*p$ < .05). (b) Measurements from males used in Experiment A performed in the laboratory (N = 27 for amplexant males, N = 27 for non-amplexant males, paired t test, $***p$ < .001)

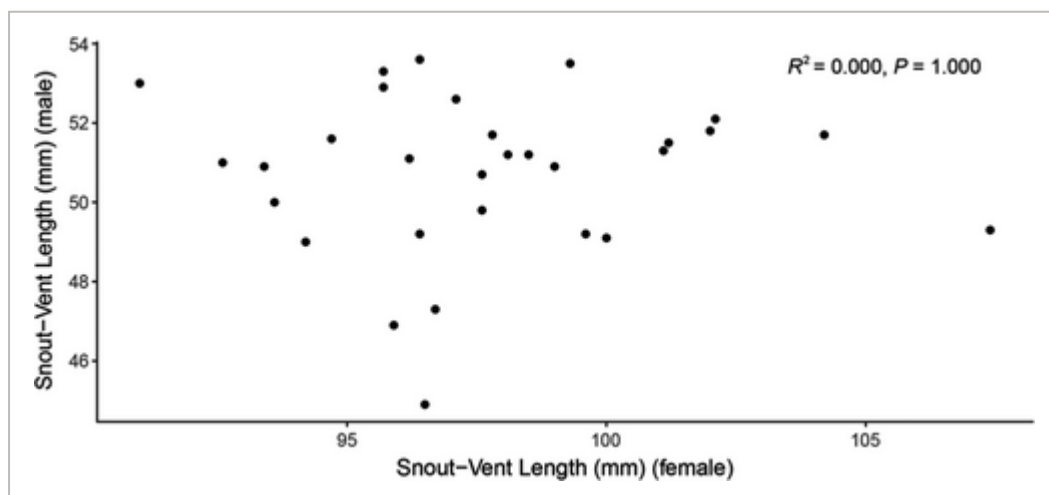


FIGURE 5

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Correlation of the snout-vent length (SVL) between amplexant males and amplexant females (N = 30, linear-regression analysis, p > .05)

3.2 Correlation between male body size and call characteristics

Analysis of the advertisement calls of 28 male frogs revealed that body size was significantly positively related to the P-NLP-C (linear-regression analysis, $R^2 = 0.120$, $p = .040$, Figure 6a), but not significantly correlated with the fundamental frequency (linear-regression analysis, $R^2 = 0.028$, $p = .216$, Figure 6b); in other words, calls of larger males showed a higher P-NLP-C, and P-NLP-C might be a potential predictor of male body size.

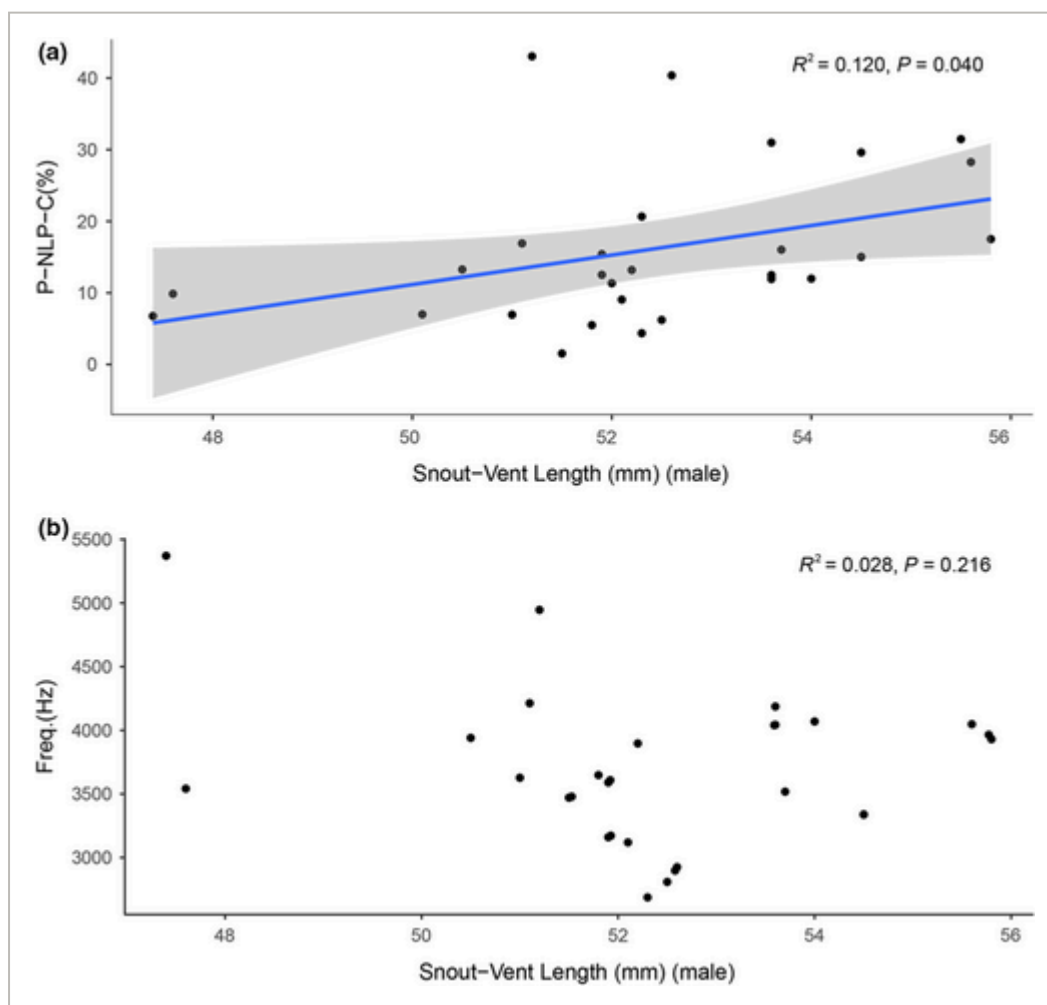


FIGURE 6

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Correlation of the snout-vent length (SVL) with the average P-NLP-C of male frogs (a) ($N = 28$, linear-regression analysis, $p < .05$) and the fundamental frequency (b) ($N = 28$, linear-regression analysis, $p > .05$). The gray areas represent the 95% confidence intervals

3.3 Female mate choice in amplexus experiments

In Experiment A, 30 females were subjected to the amplexus experiment; 3 females did not react, and 4 females chose smaller males; accordingly, 23 females indicated a preference for

larger males. Based on all the video recordings, we confirmed the general pattern of *O. graminea* mate choice. After acclimating to the test terrarium, first, both males would gradually move closer to the female, accompanied by intermittent calls; then, the female would listen carefully and observe the two male competitors for a few minutes. After much deliberation, she would adjust her body direction and turn her back toward the favorite male. Finally, this male would call again in response to the female's choice, then jumped on the female's back to complete the amplexus. The mean SVL of the amplexant males was 50.53 mm (± 1.64 SD, $N = 27$)—this differed significantly (paired t test, $p < .001$) from that of non-amplexant males (mean $\pm$ SD = 48.41 ± 1.82 mm, $N = 27$) (Figure 4b). In Experiment B, we placed a new female with two males that had been previously paired with a different female. The results showed that larger males were once again chosen by all females ($N = 20$, 100%).

3.4 Female phonotaxis

A total of 24 gravid female *O. graminea* were subjected to the phonotaxis experiments, and some females did not make a choice ($N = 4$); however, most female frogs preferred advertisement calls with a low f_0 over calls with a high f_0 (binomial test, $p = .003$, Figure 7b). Female frogs preferred advertisement calls with a high P-NLP-C (50%) compared to the advertisement calls with a low P-NLP-C (10%) (binomial test, $p = .003$, Figure 7c). Twenty gravid female frogs that responded in phonotaxis test 2 were subjected to phonotaxis test 3. The results of the experiment showed that calls with a high P-NLP-C (75%) were chosen by 18 females, while 2 females chose calls with a low P-NLP-C (20%) (binomial test, $p < .001$, Figure 7d).

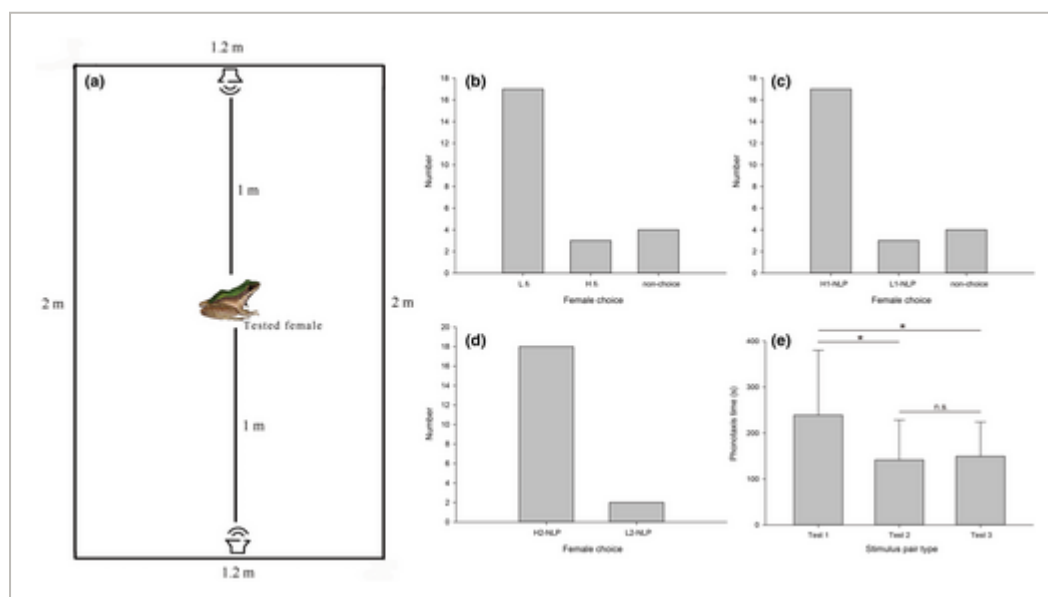


FIGURE 7

The results of female frog phonotaxis experiments. (a) Demonstration of experimental operation. (b) Female choice of test 1 ($N = 24$). (c) Female choice of test 2 ($N = 24$). (d) Female choice of test 3 ($N = 20$). (e) Difference of female phonotaxis times among different stimulus pairs ($N = 20$, Mann–Whitney U test, $*p < .05$; n.s. not statistically significant)

Furthermore, Mann–Whitney U tests revealed that female phonotaxis times among different stimulus pairs were significantly different (Figure 7e). The female phonotaxis time was significantly longer for the stimulus pairs of $L f_0$ vs $H f_0$ (mean $\pm$ SD = 238.95 ± 141.38 s) compared to that of L1-NLP vs H1-NLP (mean $\pm$ SD = 141.50 ± 87.02 s) (Mann–Whitney U test, $p < .05$) and that of L2-NLP vs H2-NLP (mean $\pm$ SD = 149.40 ± 74.90 s) (Mann–Whitney U test, $p < .05$). However, there was no significant difference in female phonotaxis time between test 2 and test 3 (Mann–Whitney U test, $p = .66$).

4 DISCUSSION

The results of our study in *O. graminea* support the working hypothesis that females would prefer males producing calls with a higher P-NLP-C. We found that in the field, females of *O. graminea* preferentially mated with larger males whose calls had a higher P-NLP-C. In manipulative experiments, females also showed a preference for males with larger body sizes and calls with a high P-NLP-C. The present study highlights the important role of NLP in anuran mate selection.

Anuran mating patterns are often nonrandom in terms of body size (Mansouri et al., 2020), which was also demonstrated in this study. Females of *O. graminea* preferentially mate with larger males in the field, which was validated in two two-choice experiments. For most anuran species, females show a preference for larger males (Andersson, 1994; Rausch et al., 2014). Mating with larger males could result in indirect genetic benefits in terms of high offspring fitness. Márquez reported that midwife toads (*Alytes obstetricans* and *A. cisternasii*) who mate with larger males produce offspring with faster growth, a larger adult size, and superior survival ability (Márquez, 1995). But there are a few exceptions, for example, female serrate-legged small treefrogs (*Philautus odontotarsus*) favored the intermediate size of males to reduce energy consumption for carrying male (Zhu et al., 2016). In addition, the body size ratio of amplexed pairs could affect the fertilization rate, for example, effective fertilization requires that cloaca to be properly juxtaposed during amplexus (Bastos & Haddad, 1996; Friedl & Klump, 2005). In *O. graminea*, there is greater female-biased sexual size dimorphism between sexes (sexual dimorphism index = 1.93, $N = 30$; Figure S1) compared to most anuran species (Monnet & Cherry, 2002; Zhang & Lu, 2013); thus, choosing a large mate is conducive to improving the efficiency of amplexus and fertilization in *O. graminea*.

Due to individual differences in vocal apparatus, body size variation can affect the occurrence or intensity of NLP components in acoustic signals and increase the complexity of vocalizations (Cazau et al., 2016; Serrano et al., 2020). In *O. graminea*, a larger body size

was associated with a higher P-NLP-C. The phonotaxis experiments indicated that the females preferred male calls with a higher P-NLP-C, so it is reasonable to speculate that a higher P-NLP-C in calls might enhance male attractiveness, affecting female mate choice and contributing to female preference for large males. It is likely that call features such as nonlinearities encode relevant information, such as body size (Juola & Searcy, 2011; Wu et al., 2021). The relationship between body size and NLP has been previously reported to occur in mammals (Cazau et al., 2016; Fitch et al., 2002), but it has not been corroborated by a significant correlation in any living organism. Serrano et al. (2020) first revealed that body size had influences on NLP components at the intra- and interpopulation levels, for example, in Darwin's frogs (*Rhinoderma darwinii*), smaller individuals had higher proportions of relative duration of chaos. This study demonstrated that there was a significantly positive correlation between the P-NLP-C in calls and body size in male *O. graminea*. An inverse relationship between body size and fundamental frequency occurs typically in many anurans, but this study did not find the significant correlation in male *O. graminea*. It may be limited to the sample size of our study and require further experimental verification. Meanwhile, the evolution of female preference for male calls with different frequencies needs to take into account the multiple effects of auditory sensitivity and background context (Zhao et al., 2017). We speculate that males of *O. graminea* with bigger body size may not always have the calls with lower f_0 in the field, otherwise, their opportunities of being detected by females would be reduced due to the masking of background stream noise. This suggests that for female *O. graminea*, the P-NLP-C may be a more reliable clue of male body size than the fundamental frequency. Further, this study also revealed that a higher P-NLP-C in courtship calls can enhance female attention to calling males, and larger males with a higher P-NLP-C had greater mating success than smaller rivals.

The complex structure and irregular frequency spectrum of NLP can increase the specificity of calls, making it easy to attract the attention of receivers and prevent habituation (Blumstein & Recapet, 2009). The results of phonotaxis experiments indicated that females of *O. graminea* had a strong preference for the calls with a high P-NLP-C or a low f_0 . However, the time of phonotaxis in the high-low P-NLP-C tests was significantly less than that in the high-low f_0 test. This suggests that a high P-NLP-C might play a more important role in female phonotactic behavior than a low f_0 , and a higher P-NLP-C in calls has stronger effect on mate choice by females. These results are in accordance with recent findings in mammals, in which females displayed the strongest preference for calls with NLP over those without NLP (Charlton et al., 2017, 2018; Reby & Charlton, 2012). Therefore, the hypothesis that calls with nonlinear components are more attractive to females is not limited to mammals, as it also applies to other animals, including frogs.

The communicative significance of NLP has been examined in anurans (Feng, Arch, et al., 2009), reptiles (Labra et al., 2013), birds (Digby et al., 2014), and mammals (Tyson et al., 2007). Different functions have been assigned to NLP, including individual recognition (Feng, Riede, et al., 2009; Fitch et al., 2002), prevention of habituation (Karp et al., 2014), indicators

of fitness (Fitch et al., 2002; Riede et al., 2004), and so on. However, the role of NLP in social interactions has rarely been evaluated (Digby et al., 2014). The study in *O. tormota* showed that male frogs with a high P-NLP-C in calls had a greater advantage in mate selection (Wu et al., 2021), consistent with the results of this study, indicating that the P-NLP-C in calls might play an important role in mate selection in frogs that can produce calls with NLP.

ACKNOWLEDGMENTS

We thank Albert S Feng for his constructive suggestions on the field and laboratory study, and always cherish the memory of this sincere mentor and friend. We also thank anonymous referees for their valuable comments which help us greatly improve the manuscript.

CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

AUTHOR CONTRIBUTION

Pan Chen: Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Jinmei Wang:** Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Visualization (equal); Writing – original draft (equal). **Junqi Miao:** Formal analysis (equal); Investigation (supporting); Methodology (supporting); Visualization (supporting). **Hao Dong:** Formal analysis (supporting); Investigation (equal). **Jiahui Bao:** Investigation (supporting); Methodology (supporting). **Yatao Wu:** Formal analysis (supporting); Investigation (supporting); Methodology (supporting). **Fang Zhang:** Conceptualization (lead); Funding acquisition (lead); Methodology (lead); Project administration (lead); Resources (lead); Supervision (lead); Writing – review & editing (lead).

DATA AVAILABILITY STATEMENT

The raw data used to perform analyses and generate figures for this manuscript are available at <https://doi.org/10.5061/dryad.sqv9s4n5h>; <https://doi.org/10.17632/xtrf2dxv6j.1>.

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Andersson, M. (1994). *Sexual selection*. Princeton University Press.

[CAS](#)  | [Google Scholar](#) 

Bastos, R. P., & Haddad, C. F. B. (1996). Breeding activity of the neotropical treefrog *Hyla elegans* (Anura, Hylidae). *Journal of Herpetology*, 30, 355–360. <https://doi.org/10.2307/1565172>

[Web of Science®](#)  | [Google Scholar](#) 

Blumstein, D. T., & Recapet, C. (2009). The sound of arousal: The addition of novel non-linearities increases responsiveness in Marmot alarm calls. *Ethology*, 115(11), 1074–1081. <https://doi.org/10.1111/j.1439-0310.2009.01691.x>

[Web of Science®](#)  | [Google Scholar](#) 

Boersma, P., & Weenink, D. (2007). PRAAT: Doing phonetics by computer (Version 5.2.22). Retrieved from <http://www.fon.hum.uva.nl/praat/>

[Google Scholar](#) 

Cazau, D., Adam, O., Aubin, T., Laitman, J. T., & Reidenberg, J. S. (2016). A study of vocal nonlinearities in humpback whale songs: from production mechanisms to acoustic analysis. *Scientific Reports*, 6, <https://doi.org/10.1038/srep31660>

[PubMed](#)  | [Web of Science®](#)  | [Google Scholar](#) 

Charlton, B. D., Martin-Wintle, M. S., Owen, M. A., Zhang, H., & Swaisgood, R. R. (2018). Vocal behaviour predicts mating success in giant pandas. *Royal Society Open. Science*, 5(10), <https://doi.org/10.1098/rsos.181323>

[Web of Science®](#)  | [Google Scholar](#) 

Charlton, B. D., Watchorn, D. J., & Whisson, D. A. (2017). Subharmonics increase the auditory impact of female koala rejection calls. *Ethology*, 123(8), 571–579. <https://doi.org/10.1111/eth.12628>

Chen, B. H. (1991). *Anhui Amphibia and Reptilia (In Chinese)*. Anhui press of Sciences and Technology.

Cui, J., Song, X., Zhu, B., Fang, G., Tang, Y., & Ryan, M. J. (2016). Receiver discriminability drives the evolution of complex sexual signals by sexual selection. *Evolution*, 70(4), 922–927. <https://doi.org/10.1111/evo.12889>

Digby, A., Bell, B. D., & Teal, P. D. (2014). Non-linear phenomena in little spotted kiwi calls. *Bioacoustics-the International Journal of Animal Sound and Its Recording*, 23(2), 113–128. <https://doi.org/10.1080/09524622.2013.829755>

Fee, M. S., Shraiman, B., Pesaran, B., & Mitra, P. P. (1998). The role of nonlinear dynamics of the syrinx in the vocalizations of a songbird. *Nature*, 395(6697), 67–71. <https://doi.org/10.1038/25725>

Feng, A. S., Arch, V. S., Yu, Z., Yu, X.-J., Xu, Z.-M., & Shen, J.-X. (2009). Neighbor-stranger discrimination in concave-eared Torrent Frogs, *Odorrana tormota*. *Ethology*, 115(9), 851–856. <https://doi.org/10.1111/j.1439-0310.2009.01666.x>

Feng, A. S., Riede, T., Arch, V. S., Yu, Z., Xu, Z.-M., Yu, X.-J., & Shen, J.-X. (2009). Diversity of the Vocal Signals of Concave-Eared Torrent Frogs (*Odorrana tormota*): Evidence for Individual Signatures. *Ethology*, 115(11), 1015–1028. <https://doi.org/10.1111/j.1439-0310.2009.01692.x>

Fitch, W. T., Neubauer, J., & Herzel, H. (2002). Calls out of chaos: the adaptive significance of nonlinear phenomena in mammalian vocal production. *Animal Behaviour*, 63, 407–418. <https://doi.org/10.1006/anbe.2001.1912>

[Web of Science®](#) [Google Scholar](#)

Friedl, T. W. P., & Klump, G. M. (2005). Sexual selection in the lek-breeding European treefrog: body size, chorus attendance, random mating and good genes. *Animal Behaviour*, **70**, 1141–1154. <https://doi.org/10.1016/j.anbehav.2005.01.017>

[Web of Science®](#) [Google Scholar](#)

Gridi-Papp, M., Rand, A. S., & Ryan, M. J. (2006). Animal communication: Complex call production in the tungara frog. *Nature*, **441**(7089), 38. <https://doi.org/10.1038/441038a>

[CAS](#) [PubMed](#) [Web of Science®](#) [Google Scholar](#)

Juola, F. A., & Searcy, W. A. (2011). Vocalizations reveal body condition and are associated with visual display traits in great frigatebirds (*Fregata minor*). *Behavioral Ecology and Sociobiology*, **65**(12), 2297–2303. <https://doi.org/10.1007/s00265-011-1240-0>

[Web of Science®](#) [Google Scholar](#)

Kaplan, J. D., Melillo-Sweeting, K., & Reiss, D. (2018). Biphonal calls in Atlantic spotted dolphins (*Stenella frontalis*): bitonal and burst-pulse whistles. *Bioacoustics-the International Journal of Animal Sound and Its Recording*, **27**(2), 145–164. <https://doi.org/10.1080/09524622.2017.1300105>

[Web of Science®](#) [Google Scholar](#)

Karp, D., Manser, M. B., Wiley, E. M., & Townsend, S. W. (2014). Nonlinearities in Meerkat alarm calls prevent receivers from habituating. *Ethology*, **120**(2), 189–196. <https://doi.org/10.1111/eth.12195>

[Web of Science®](#) [Google Scholar](#)

Labra, A., Silva, G., Norambuena, F., Velasquez, N., & Penna, M. (2013). Acoustic features of the weeping Lizard's distress call. *Copeia*, **2013**(2), 206–212. <https://doi.org/10.1643/CE-12-026>

[Web of Science®](#) [Google Scholar](#)

Mansouri, I., Dakki, M., Squalli, W., Ousaaid, D., Elfalah, S., & Elghadraoui, L. (2020). Field investigation of Turtle doves' courtship: vocal calls versus arc-flight. *Journal of Animal Behaviour and Biometeorology*, **8**(1), 32–40. <https://doi.org/10.31893/jabb.20005>

[Web of Science®](#) [Google Scholar](#)

Márquez, R. (1995). Female Choice in the Midwife Toads (*Alytes Obstetricans* and *a. Cisternasii*). *Behaviour*, 132(1–2), 151–161. <https://doi.org/10.1163/156853995X00342>

[Web of Science®](#) [Google Scholar](#)

Monnet, J. M., & Cherry, M. I. (2002). Sexual size dimorphism in anurans. *Proceedings of the Royal Society B-Biological Sciences*, 269(1507), 2301–2307. <https://doi.org/10.1098/rspb.2002.2170>

[PubMed](#) [Web of Science®](#) [Google Scholar](#)

Narins, P. M., Feng, A. S., Fay, R. R., & Popper, A. N. (2006). *Hearing and Sound Communication in Amphibians*. Springer.

[Google Scholar](#)

Rausch, A. M., Sztatecsny, M., Jehle, R., Ringler, E., & Hödl, W. (2014). Male body size and parental relatedness but not nuptial colouration influence paternity success during scramble competition in *Rana arvalis*. *Behaviour*, 151(12–13), 1869–1884. <https://doi.org/10.1163/1568539X-00003220>

[Web of Science®](#) [Google Scholar](#)

Rebout, N., De Marco, A., Lone, J.-C., Sanna, A., Cozzolino, R., Micheletta, J., & Thierry, B. (2020). Tolerant and intolerant macaques show different levels of structural complexity in their vocal communication. *Proceedings of the Royal Society B-Biological Sciences*, 287(1928), 20200439. <https://doi.org/10.1098/rspb.2020.0439>

[Web of Science®](#) [Google Scholar](#)

Reby, D., & Charlton, B. D. (2012). Attention grabbing in red deer sexual calls. *Animal Cognition*, 15(2), 265–270. <https://doi.org/10.1007/s10071-011-0451-0>

[PubMed](#) [Web of Science®](#) [Google Scholar](#)

Rice, A. N., Land, B. R., & Bass, A. H. (2011). Nonlinear acoustic complexity in a fish ‘two-voice’ system. *Proceedings of the Royal Society B-Biological Sciences*, 278(1725), 3762–3768. <https://doi.org/10.1098/rspb.2011.0656>

[PubMed](#)  | [Web of Science®](#)  | [Google Scholar](#) 

Riede, T., Owren, M. J., & Arcadi, A. C. (2004). Nonlinear acoustics in pant hoots of common chimpanzees (*Pan troglodytes*): Frequency jumps, subharmonics, biphonation, and deterministic chaos. *American Journal of Primatology*, 64(3), 277–291. <https://doi.org/10.1002/ajp.20078>

[CAS](#)  | [PubMed](#)  | [Web of Science®](#)  | [Google Scholar](#) 

Ryan, M. J. (1983). Sexual selection and communication in a neotropical frog, *Physalaemus pustulosus*. *Evolution*, 37(2), 261–272. <https://doi.org/10.1111/j.1558-5646.1983.tb05536.x>

[PubMed](#)  | [Web of Science®](#)  | [Google Scholar](#) 

Ryan, M. J., Page, R. A., Hunter, K. L., & Taylor, R. C. (2019). ‘Crazy love’: nonlinearity and irrationality in mate choice. *Animal Behaviour*, 147, 189–198. <https://doi.org/10.1016/j.anbehav.2018.04.004>

[Web of Science®](#)  | [Google Scholar](#) 

Serrano, J. M., Penna, M., & Soto-Azat, C. (2020). Individual and population variation of linear and non-linear components of the advertisement call of Darwin's frog (*Rhinoderma darwinii*). *Bioacoustics-the International Journal of Animal Sound and Its Recording*, 29(5), 572–589. <https://doi.org/10.1080/09524622.2019.1631214>

[Web of Science®](#)  | [Google Scholar](#) 

Shen, J.-X., Xu, Z.-M., Feng, A. S., & Narins, P. M. (2011). Large odorous frogs (*Odorrana graminea*) produce ultrasonic calls. *Journal of Comparative Physiology a-Neuroethology Sensory Neural and Behavioral Physiology*, 197(10), 1027–1030. <https://doi.org/10.1007/s00359-011-0660-7>

[PubMed](#)  | [Web of Science®](#)  | [Google Scholar](#) 

Townsend, S. W., & Manser, M. B. (2011). The function of nonlinear phenomena in meerkat alarm calls. *Biology Letters*, 7(1), 47–49. <https://doi.org/10.1098/rsbl.2010.0537>

[PubMed](#)  | [Web of Science®](#)  | [Google Scholar](#) 

Tyson, R. B., Nowacek, D. P., & Miller, P. J. O. (2007). Nonlinear phenomena in the vocalizations of North Atlantic right whales (*Eubalaena glacialis*) and killer whales (*Orcinus orca*). *Journal of the Acoustical Society of America*, 122(3), 1365–1373. <https://doi.org/10.1121/1.2756263>

Volodina, E. V., Volodin, I. A., Isaeva, I. V., & Unck, C. (2006). Biphonation may function to enhance individual recognition in the dhole, *Cuon Alpinus*. *Ethology*, 112(8), 815–825. <https://doi.org/10.1111/j.1439-0310.2006.01231.x>

Wilden, I., Herzel, H., Peters, G., & Tembrock, G. (1998). Subharmonics, biphonation, and deterministic chaos in mammal vocalization. *Bioacoustics*, 9(3), 171–196. <https://doi.org/10.1080/09524622.1998.9753394>

Wu, Y., Bao, J., Lee, P., Wang, J., Wang, S., & Zhang, F. (2021). Nonlinear phenomena conveying body size information and improving attractiveness of the courtship calls in the males of *Odorrana tormota*. *Asian Herpetological Research*, 12(1), 117–123. <https://doi.org/10.16373/j.cnki.ahr.200028>

Zhang, F., Chen, P., Chen, Z., & Zhao, J. (2015). Ultrasonic frogs call at a higher pitch in noisier ambiance. *Current Zoology*, 61(6), 996–1003. <https://doi.org/10.1093/czoolo/61.6.996>

Zhang, F., Wu, Y., Wang, J., & Bao, J. (2021). Calls of the large Odorous Frog (*Odorrana graminea*) show nonlinear and individual characteristics. *Asian Herpetological Research*, 12(1), 124–134. <https://doi.org/10.16373/cnki.ahr.200070>

Zhang, F., Yuan, C., & Feng, A. S. (2020). Female concave-eared torrent frogs prefer smaller males. *Journal of Zoology*, 311(4), 239–245. <https://doi.org/10.1111/jzo.12775>

Zhang, F., Zhao, J., & Feng, A. S. (2017). Vocalizations of female frogs contain nonlinear characteristics and individual signatures. *PLoS One*, 12(3), <https://doi.org/10.1371/journal.pone.0174815>

Zhang, L., & Lu, X. (2013). Sexual size dimorphism in anurans: Ontogenetic determination revealed by an across-species comparison. *Evolutionary Biology*, 40(1), 84–91. <https://doi.org/10.1007/s11692-012-9187-2>

Zhao, L., Zhu, B., Wang, J., Brauth, S. E., Tang, Y., & Cui, J. (2017). Sometimes noise is beneficial: stream noise informs vocal communication in the little torrent frog *Amolops torrentis*. *Journal of Ethology*, 35(3), 259–267. <https://doi.org/10.1007/s10164-017-0515-y>

Zhu, B., Wang, J., Zhao, L., Chen, Q., Sun, Z., Yang, Y., & Cui, J. (2017). Male-male competition and female choice are differentially affected by male call acoustics in the serrate-legged small treefrog, *Kurixalus odontotarsus*. *PeerJ*, 5, e3980. <https://doi.org/10.7717/peerj.3980>

Zhu, B., Wang, J., Zhao, L., Sun, Z., Brauth, S. E., Tang, Y., & Cui, J. (2016). Bigger is not always better: Females prefer males of mean body size in *Philautus odontotarsus*. *PLoS One*, 11(2), e0149879. <https://doi.org/10.1371/journal.pone.0149879>

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